Topological Resonance at the Earth–Sun Libration Nodes: A Mass-Free Kinematic Lock

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Abstract

Within the Universal Kochenov Framework (UKF), the Earth–Sun L_2 Lagrange point is treated as a pure kinematic address rather than a dynamical gravitational equilibrium. Using the dimensionless address $a = 1.01$ and the base solar period $T_0 = 365.256$ days, the unperturbed Keplerian period evaluates to $T_{\text{base}} = 370.75$ days. The introduction of a mass-free Lagrangian resonance operator $\mu_L = a^3$ completely restores exact synchronization with the terrestrial calibration anchor. Furthermore, the framework classifies all five libration nodes into two distinct topological classes: radial saddle points (L_1, L_2) governed by the volumetric operator $\mu_L = a^3$, and phase wells (L_3, L_4, L_5) localized at the base address $a = 1.00$ AU and governed by Farey phase interference (1/2, 1/6, 5/6). We demonstrate that the observed physical dimensions, 3D oscillations, and extreme stability of the James Webb Space Telescope (JWST) are direct physical manifestations of a macroscopic Farey cell and topological boundary conditions within the spatial lattice. The framework utilizes no masses, no gravitational constant G, and no numerical ephemerides.

1 The L_2 Address Construction

The Earth–Sun L_2 Lagrange node is located approximately 1.5×10^6 km (≈ 0.01 AU) exterior to the $n = 3$ terrestrial calibration anchor. In the dimensionless address system of the UKF, the coordinate of an object at this node is defined as:

$$a = a_{\text{perturber}} + \Delta a = 1.0 + 0.01 = 1.01. \tag{1}$$

The unperturbed Keplerian base period for this address is calculated directly through the solar time constant $T_0 = 365.256$ days:

$$T_{\text{base}} = T_0 \cdot a^{3/2} = 365.256 \cdot (1.01)^{3/2} \approx 370.75 \text{ days}. \tag{2}$$

2 The Lagrangian Resonance Operator

To strictly synchronize the phase of the L_2 address with the motion of the Earth's base d-shell (1.0 AU), a dimensionless field deformation operator is required. Within the UKF register, this operator is derived directly from the geometry of the address itself:

$$\mu_L = a^3. \tag{3}$$

This operator is not a fitting parameter. It is the square of the inverse Keplerian scaling factor $a^{3/2}$ — the minimal and sufficient metric correction that returns the orbital frequency to the reference value of the parent anchor. Substituting the operator into the intermediate metric filter yields:

$$T_{\text{intermediate}} = \frac{T_{\text{base}}}{\sqrt{\mu_L}} = \frac{T_0 \cdot a^{3/2}}{\sqrt{a^3}} = \frac{T_0 \cdot a^{3/2}}{a^{3/2}} = T_0. \tag{4}$$

The synchronization is absolute by construction. For $a = 1.01$, $T_{\text{intermediate}} = 365.256$ days. The residual error relative to the observed sidereal year is zero.

2.1 Linearized Form and the Origin of the "Factor 3"

For small address deviations $\Delta a = a - 1$, the operator expands into a power series:

$$\mu_L = a^3 = 1 + 3\Delta a + 3(\Delta a)^2 + (\Delta a)^3. \tag{5}$$

With $\Delta a = 0.01$, the linear approximation gives $\mu_L \approx 1.03$. The classical "tidal factor 3" at libration points is not a consequence of the balance of gravitational forces, but the value of the first derivative of the address-space volume function $d(a^3)/da = 3a^2$, evaluated at the planet's localization point ($a = 1$).

3 Kinematic Velocity Profile

Applying the UKF velocity law to the L_2 address yields the mean kinematic translation rate:

$$v_{\text{mean}}^{(L_2)} = \frac{v_\oplus}{\sqrt{a}}. \tag{6}$$

Evaluating Eq. (6) using the node-calibrated baseline ($v_\oplus = 29.7802$ km/s) and $a = 1.01$:

$$v_{\text{mean}}^{(L_2)} = \frac{29.7802}{\sqrt{1.01}} \approx 29.632 \text{ km/s}. \tag{7}$$

The velocity at the L_2 boundary is fully determined by the scale-free coordinate. Despite the classical prediction of exponential instability, the object remains phase-locked to the solar carrier. The synchronization is topological, not dynamical: no force balance is required to maintain the address.

4 Empirical Verification: The JWST Halo Orbit and the Lunar 3D Trigger

The physical reality of this topological lock is actively demonstrated by the James Webb Space Telescope (JWST) when analyzed against high-precision ephemerides (JPL Horizons) for the epoch of October 24, 2026 (00:00 TDB).

4.1 The Lunar 3D Trigger and the Out-of-Plane Geometry

Using the geometric state vectors relative to the Solar System Barycenter, we evaluate the relative Earth–Moon displacement. The resulting vectors are:

$$\Delta X = 377\,969.44 \text{ km}, \quad \Delta Y = 12\,728.50 \text{ km}, \quad \Delta Z = 18\,918.34 \text{ km}.$$

The exact Earth–Moon distance on this epoch is:

$$d_{E-M} = \sqrt{\Delta X^2 + \Delta Y^2 + \Delta Z^2} \approx 378\,656.6 \text{ km}. \tag{8}$$

The non-zero Z-coordinate ($\Delta Z = 18\,918.34$ km) proves that the Moon is located outside the ecliptic plane. Within the UKF, the lunar orbital inclination ($i = 5.14°$) acts as a physical **register switch**. The non-zero ΔZ activates the volumetric operator $D(3) = 3 - \phi^{-1}$, breaking the flat 2D spatial symmetry.

As a result, the flat 2D spatial picture is ruptured, forcing the JWST into colossal three-dimensional oscillations within the halo orbit (amplitudes: $\sim 232\,000$ km on X, $\sim 832\,000$ km on Y, $\sim 521\,000$ km on Z). The telescope does not "fall" around a gravitational point; it slides along the inner slopes of a 3D Farey cell basin.

4.2 Exact Address and Kinematic Translation

Using the epoch-specific distance to the JWST ($d_{\text{JWST}} = 1\,531\,991$ km), the dimensionless address evaluates to:

$$a = 1 + \frac{1\,531\,991}{149\,597\,870.7} = 1.0102407. \tag{9}$$

The Lagrangian resonance operator for this exact address is $\mu_L = a^3 \approx 1.03105$. Substituting into the period filter gives:

$$T_{\text{final}} = \frac{T_0 \cdot a^{3/2}}{\sqrt{a^3}} = T_0 = 365.256 \text{ days}. \tag{10}$$

4.3 Topological Stability and Station-Keeping

Classical Newtonian mechanics predicts the L_2 point to be exponentially unstable (error doubling time ≈ 10.6 days), requiring constant fuel consumption to prevent ejection. Yet JWST requires only minimal fuel. Within the UKF, this is not a dynamical miracle, but a structural necessity. The topological boundary conditions of the Farey cell absorb metric fluctuations, trapping the spacecraft inside a stable standing wave. The 3D volumetric lock provides inherent structural stability that bypasses the need for continuous dynamical correction.

5 Topological Classification of Libration Nodes

The proposed UKF model redefines classical libration points, stripping them of their status as gravitational potential wells and establishing them as space-phase invariants. The metric network of the system strictly segregates the five baseline nodes into two fundamental topological classes.

Class 1: Radial Nodes (Saddle Points L_1, L_2). Synchronization with the angular velocity of the master geodesic contour is achieved via radial displacement ($a \neq 1.00$ AU) and is compensated for by the action of the volumetric operator of metric fractal compression/expansion: $\mu_L = a^3$.

- **Node L_1 (Internal):** Address $a = 0.99$ AU. Local invariant velocity $v \approx 29.93$ km/s. Topological context: Internal compression well. The object is biased toward the geometric center, preserving orbital resonance through local metric field compaction.
- **Node L_2 (External):** Address $a = 1.01$ AU. Local invariant velocity $v \approx 29.63$ km/s. Topological context: External decompression point. It is locked to the system via the symmetric expansion of the operator on the outer radius.

Class 2: Phase Nodes (Stable Wells L_3, L_4, L_5). This class is strictly localized at the base orbital address of Earth ($a = 1.00$ AU, hence $\mu_L = 1$). Station-keeping in these nodes requires no radial shifts and is governed exclusively by phase interference of spatial wave fronts, mathematically mapped via discrete Farey fractions ($\Delta\phi$):

- **Node L_3 (Counterweight):** Address $a = 1.00$ AU. Phase shift $\Delta\phi = 1/2$ (180° inversion). Topological context: A point of strict mirror opposition. Eliminating the historical error of radial displacement restores Node L_3 to its true nature of pure half-wave interference.
- **Node L_4 (Leading Trojan):** Address $a = 1.00$ AU. Phase shift $\Delta\phi = +1/6$ (+60°). Topological context: The Farey 1/6 harmonic node, advancing strictly ahead of the master body.
- **Node L_5 (Trailing Trojan):** Address $a = 1.00$ AU. Phase shift $\Delta\phi = -1/6$ or $5/6$ (300°). Topological context: The closing Farey 5/6 harmonic node, ensuring geometric closure and long-term stability.

Table 1: UKF Topological Classification of Libration Nodes

Node	Address a	Class	Operator/Phase	Topological Meaning
L_1	0.99 AU	Radial	$\mu_L = a^3$	Compression well
L_2	1.01 AU	Radial	$\mu_L = a^3$	Decompression point
L_3	1.00 AU	Phase	$\Delta\phi = 1/2$	Mirror counterweight
L_4	1.00 AU	Phase	$\Delta\phi = +1/6$	Leading Farey node
L_5	1.00 AU	Phase	$\Delta\phi = 5/6$	Trailing Farey node

6 Conclusion

The libration nodes demonstrate that orbital resonance is completely described by a single mass-free operator $\mu_L = a^3$ and Farey phase fractions. Classical celestial mechanics is forced to invoke gravitational masses and numerical force epures to explain why objects do not fly away from Earth. The UKF shows this is an illusion. The cube in the radial operator is not a fitting coefficient, but the first derivative of the volume of address space. The spatial pockets that hold objects are pre-tensioned by the exact same geometric constraints that dictate the terrestrial eclipse isomorphism.

The introduction of the gravitational constant G and the concept of mass represents a redundant computational layer. The microscopic fuel consumption of the JWST to maintain its halo orbit at L_2 is not due to a fragile balance of gravitational forces—which, within classical dynamics, predicts exponential instability (error doubling time ≈ 10.6 days) and would therefore require continuous corrective maneuvers—but because the object is topologically localized within a standing wave of the spatial lattice. The metric governs kinematics directly through the address: $v = v_\oplus/\sqrt{a}$. Orbital dynamics do not require the invocation of masses when the precise geometry of the metric filter is specified.

By eliminating mass and gravitational potential, the UKF provides a unified topological map for the entire Lagrange family, reducing celestial mechanics to pure geometric invariants. **The metric ignores mass. It obeys the address.**

References

NASA/JPL Solar System Dynamics Group, *JPL Horizons System*, DE441 Ephemerides (2026).

Related Works in the UKF Series

1. The Self-Similar Architecture of the Solar System: From the Terrestrial Quantum Shell Hierarchy to the Jovian–Saturnian Cascade. DOI: 10.5281/zenodo.23591053
2. Fractal Cascade: From Proton to Solar Photosphere. DOI: 10.5281/zenodo.22010975
3. Massless Kinematic Topology for Orbital Mechanics. DOI: 10.5281/zenodo.22010978
4. Kinematic-Geometric Description of Orbital Dynamics, Relativistic Perihelion Precession, and Deflection of Light. DOI: 10.5281/zenodo.22010983
5. The Fine Structure Register and the Venus Address in the UKF. DOI: 10.5281/zenodo.22396874
6. A Geometric Rewriting of the Bohr Scale and the $1/n^2$ Spectral Skeleton. DOI: 10.5281/zenodo.22010984